in marine investigations as well as in teeth, tracking animal migrations or in criminal forensics.

3. Dating of rocks.

4. As a radioactive tracer in determining the source of ancient archaeological materials such as timbers and coins.

Uses of Barium

1. Used in metallurgy, its compounds are used in pyrotechnics, petroleum mining and radiology.

2. Deoxidiser in copper refining.

3. Its alloys with nickel readily emits electrons hence used in electron tubes and in spark plug electrodes.

4. As a scavenger to remove last traces of oxygen and other gases in television and other electronic tubes.

5. An isotope of barium ^{133}Ba, used as a source in the calibration of gamma ray detectors in nuclear chemistry.

Uses of Radium

Used in self-luminous paints for watches, nuclear panels, aircraft switches, clocks and instrument dials.

5.6. General characteristics of the compounds of the alkaline earth metals

The dipositive oxidation state (M^{2+}) is the predominant valence of group 2 elements. The alkaline earth metals form compounds which are predominantly ionic. However, they are less ionic than the corresponding compounds of alkali metals. This is due to increased nuclear charge and smaller size. The general characteristics of some of the compounds of alkaline earth metals are described below.

(a) Oxides

Generally alkaline earth metals form monoxides and peroxides.

Monoxides

Monoxides are obtained by heating the metals in oxygen. BeO and MgO are almost insoluble in water. On the other hand, oxides of other elements form hydroxides. BeO is amphoteric; MgO is weakly basic while CaO, SrO and BaO are strongly basic.

BeO oxide is covalent due to the small size of Be^{2+}ion,while other oxides are ionic in nature.

Peroxides

Except beryllium, all the remaining metals form peroxides. It is prepared by heating monoxides with oxygen at high temperature.

$$2\ BaO + O_2 \longrightarrow 2\ BaO_2$$

b)Hydroxides:

All the oxides except BeO are basic in nature and react with water to form sparingly soluble hydroxides.

$$MO + H_2O \rightarrow M(OH)_2$$

The solubility, thermal stability

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